

TASK 1: Language Translation Tool using Machine Learning

Git repository link: https://github.com/rooparooopa1029-web/CodeAlpha__LanguageTranslationTool

Aim

To translate text from one language to another using Machine Learning and Natural Language Processing (NLP).

The project uses a pretrained translation model to convert input text into the selected target language.

Introduction

Language Translation Tool is a Natural Language Processing (NLP) project.

This application translates text between different languages such as:

- English → Hindi
- English → French
- English → Spanish

using Machine Learning models.

The project helps in understanding:

- Natural Language Processing (NLP)
- Transformer Models
- Text Translation
- Flask Web Applications
- Machine Learning Integration

Tools and Libraries Used

- Python
- Flask
- Transformers
- Torch
- SentencePiece
- HTML
- CSS
- SQLite Database

Official library websites:

- [Flask](#)
- [Hugging Face Transformers](#)

Dataset Description

The project uses pretrained translation datasets and models from Hugging Face.

Translation model used:

Helsinki-NLP/opus-mt-en-hi

This model translates:

English → Hindi

Example:

Input: How are you?

Output: आप कैसे हैं?

Machine Learning Steps

1. Import libraries
2. Load pretrained translation model
3. Create Flask web application
4. Accept user input
5. Translate text using NLP model
6. Display translated text
7. Store translations in database

Python Program

```
from flask import Flask, render_template, request
from transformers import MarianMTModel, MarianTokenizer
import sqlite3

app = Flask(__name__)

# Translation model
model_name = "Helsinki-NLP/opus-mt-en-hi"

# Load tokenizer and model
tokenizer = MarianTokenizer.from_pretrained(model_name)
model = MarianMTModel.from_pretrained(model_name)

# Create database
def init_db():

    conn = sqlite3.connect('translations.db')
    cursor = conn.cursor()

    cursor.execute('''
CREATE TABLE IF NOT EXISTS translations (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    input_text TEXT,
```

```

        translated_text TEXT
    )
    '''

    conn.commit()
    conn.close()

init_db()

@app.route("/", methods=["GET", "POST"])

def home():

    translated_text = ""

    if request.method == "POST":

        text = request.form["text"]

        # Translation
        translated = tokenizer(
            [text],
            return_tensors="pt",
            padding=True
        )

        translated_tokens = model.generate(**translated)

        translated_text = tokenizer.decode(
            translated_tokens[0],
            skip_special_tokens=True
        )

        # Save to database
        conn = sqlite3.connect('translations.db')
        cursor = conn.cursor()

        cursor.execute(
            "INSERT INTO translations (input_text, translated_text)
VALUES (?, ?)",
            (text, translated_text)
        )

        conn.commit()
        conn.close()

    return render_template(
        "index.html",
        translated_text=translated_text
    )

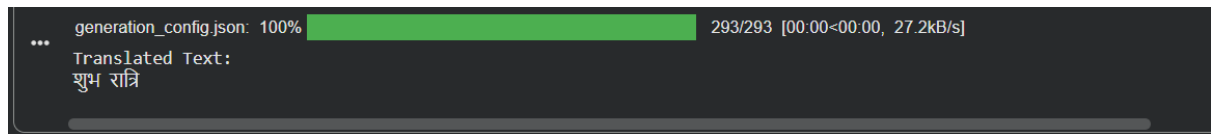
if __name__ == "__main__":
    app.run(debug=True)

```

Output:

Good Morning

सुप्रभात



Explanation of the Code:

1. Import Libraries

Required libraries for Flask, Machine Learning, and database handling are imported.

2. Load Translation Model

Pretrained Hugging Face translation model is loaded.

3. Create Database

SQLite database is created to store translations.

4. User Input

User enters text in the web interface.

5. Translation Process

The NLP model processes the text and generates translated output.

6. Store Translation

Translated text is saved into the database.

7. Display Result

Translated output is shown on the webpage.

Advantages

- Simple and beginner friendly
- Uses real Machine Learning models
- Fast translation process
- User-friendly interface
- Stores translation history

Applications

- Language learning
- Communication tools
- Chat applications
- Educational platforms
- NLP research

Conclusion

The Language Translation Tool successfully translates text using Machine Learning and NLP techniques. The project demonstrates the implementation of transformer-based language translation models along with Flask web development and database integration.